

BIEK Assistant - Dataset, Model, and Method Summary

Overview

We built a BIEK-focused local assistant for student-facing questions about the Board of Intermediate Education Karachi website. The system answers using a BIEK knowledge base built from public BIEK website pages, PDFs, forms, date sheets, model papers, notifications, and result documents.

Important clarification: we did not train a new BIEK-only model. BIEK knowledge is supplied through retrieval-augmented generation (RAG), where the model answers from indexed BIEK evidence.

Dataset

The BIEK dataset was created from public BIEK website content.

Main sources:

- BIEK HTML pages
- BIEK downloadable PDFs
- online forms
- date sheets
- model papers
- notifications and circulars
- result gazettes
- affiliated college and recognition pages
- public navigation links from the BIEK website

Extraction methods:

- HTML text extraction for website pages
- PDF text extraction with pymupdf
- OCR with tesseract_ocr where PDFs required it
- nav-tree extraction from the visible BIEK website menu

Final BIEK KB corpus:

- 967 total chunks before indexing
- 965 vectors indexed in FAISS

Chunk breakdown:

- 660 student-facing chunks
- 307 nav-backed website coverage chunks

Source material:

- 372 enriched BIEK source documents

The final indexed source file is:

```
data/runtime/biek_student_plus_nav_chunks.jsonl
```

Knowledge Base

The BIEK knowledge base is stored locally using FAISS.

Index details:

- FAISS index: `data/knowledge-base/biek/faiss.index`
- metadata: `data/knowledge-base/biek/index_meta.jsonl`
- manifest: `data/knowledge-base/biek/index_manifest.json`
- embedding model: `nomic-embed-text`
- embedding dimension: 768
- similarity metric: normalized cosine similarity

The KB includes both document chunks and nav-backed chunks. The nav-backed layer was added because important pages like Contact Us, Authorized Banks, Board Members, forms, and model papers must be reliably reachable from the public site structure.

Model

The app uses local Ollama models.

Primary runtime model:

- UI label: Synapse Qwen 2.5 1.5B V2
- Ollama tag: `synapse-1.5b-v2`
- base family: `Qwen2.5-1.5B-Instruct`
- embedding model: `nomic-embed-text`

The model generates natural answers after the BIEK KB retrieves the relevant evidence. The BIEK facts are not memorized in the model weights.

Training Method

The project contains a LoRA fine-tuning pipeline for compact local models.

For the current Synapse Qwen 1.5B V2 model:

- base model: `Qwen2.5 1.5B Instruct`
- method: LoRA fine-tuning
- framework: MLX-LM
- optimizer: Adam
- iterations: 800
- batch size: 1
- learning rate: `1e-4`
- train/validation split: 90/10
- dataset: `data/mixed_dataset_v2.jsonl`
- total samples: 558
- generated train rows: 502
- adapter output: `models/qwen1.5b-finetuned-v2`
- exported Ollama model: `synapse-1.5b-v2`

For BIEK specifically, the work was not additional fine-tuning. The BIEK implementation was:

- 1 scrape and extract public BIEK website data
- 2 enrich and classify documents
- 3 chunk student-relevant content
- 4 extract and use the public BIEK nav tree
- 5 create nav-backed chunks
- 6 merge student chunks with nav chunks
- 7 embed chunks with `nomic-embed-text`
- 8 build a FAISS index
- 9 answer with the local SLM using retrieved BIEK evidence

Runtime Flow

When a user asks a BIEK question:

- 1 The Streamlit app receives the question.
- 2 BIEK-specific routes check for direct cases such as contact, authorized banks, result lookup, forms, and nav sections.
- 3 If needed, the question is embedded with `nomic-embed-text`.
- 4 FAISS retrieves relevant BIEK chunks.
- 5 BIEK-specific reranking and guards reduce stale or unrelated evidence.
- 6 The local SLM receives the question plus retrieved evidence.
- 7 The model generates a grounded answer with links when useful.

BIEK-Specific Improvements

Key improvements added for BIEK:

- BIEK-only knowledge domain
- nav-tree coverage file
- nav-to-content mapping
- nav-backed KB chunks
- merged student + nav FAISS index
- year-aware reranking for 2026 and 2025 content
- direct handling for Contact Us and Authorized Banks
- result lookup by roll number
- shared-link handling for verification and duplicate admit card forms
- safer handling for chairman, board members, and committees
- cleanup to prevent Synapse-style wording in BIEK mode
- batch evaluation set for BIEK questions

Evaluation

The BIEK evaluation set tests realistic questions students are likely to ask:

- contact information
- authorized bank

- date sheet
- forms
- verification forms
- duplicate admit card
- model papers
- result lookup
- nav section summaries
- scheme of studies

The best BIEK-oriented run discussed during development reached about 89.8 percent accuracy on 59 questions, with an average response time around 2.82 seconds.

Final Summary

This is a local BIEK RAG assistant powered by a compact Qwen/Synapse SLM. The model provides the language generation, while the BIEK knowledge comes from a local FAISS index built from scraped and extracted BIEK website content.

This approach is better than BIEK-only fine-tuning for now because BIEK data changes frequently. Updating the KB is faster, safer, and easier to audit than retraining the model every time the website changes.